

Designing Any Imaging System from Natural Language: Agent-Constrained Composition over a Finite Primitive Basis

Chengshuai Yang

NextGen PlatformAI C Corp, USA

Correspondence: integrityyang@gmail.com

March 13, 2026

Summary. Designing a computational imaging system—selecting the forward model, validating physical feasibility, and predicting reconstruction quality—requires deep expertise in wave optics, detector physics, and modality-specific signal processing. This expertise bottleneck limits how quickly new modalities can be prototyped and contributes to the reproducibility crisis in imaging science [1]. Here we present a specification-driven agent framework that translates natural-language imaging system descriptions into validated, executable forward models with bounded design-to-real error. The framework centers on `spec.md`, a structured specification format that separates the imaging problem (*what*) from the numerical method (*how*). Three agents—Plan ($\mathcal{A}_P$), Judge ($\mathcal{A}_J$), and Performance ($\mathcal{A}_\Pi$)—generate, validate, and evaluate specifications over an 11-primitive operator basis [2]. We prove that the total design-to-real reconstruction error decomposes into five independently bounded terms (Theorem 2), and we validate the framework across 168+ imaging modalities spanning 5 carrier families and 20+ application domains—from medical CT and MRI to cryo-EM, gravitational wave detection, and neural radiance fields. On real clinical CT data (LoDoPaB [3], 200 cases), the framework achieves 98% of expert-tuned reconstruction quality with a human-to-agent efficiency ratio $\rho_{\text{time}} \approx 420$ (25 minutes vs. 7 days of expert setup). The Judge Agent rejects ill-posed specifications with 96% precision; the Performance Agent predicts reconstruction PSNR within ± 1.8 dB ($R^2 = 0.91$). Ablation studies confirm that all three agents are necessary.

1 Introduction

Computational imaging systems share a universal mathematical structure: a forward model A maps an unknown object x to measurements $y = A(x) + \text{noise}$, and reconstruction inverts this mapping [7]. Designing A for a new system requires selecting the right physical operators (Radon projection for CT, k-space encoding for MRI, coded aperture modulation for spectral imaging), setting dozens of parameters (number of angles, undersampling pattern, noise model), and validating that the composition is physically consistent and numerically well-conditioned. Today this process is entirely manual, requiring weeks of expert effort per modality.

We previously established that 11 canonical primitives—Propagate, Modulate, Project, Encode, Convolve, Accumulate, Detect, Sample, Disperse, Scatter, and Transform—suffice to represent any imaging forward model at Tier-2 fidelity across 168+ modalities and 5 carrier families [2]. This *Finite Primitive Basis* (FPB) provides the mathematical foundation but not the design tool. The question we address here is: **can an agent framework automatically design, validate, and evaluate imaging systems from natural-language descriptions, with formal guarantees on the total error from specification to real reconstruction?**

Our contributions:

1. A *specification-driven design framework* in which `spec.md`—not code—is the canonical scientific artifact, generated by a Plan Agent from natural-language input and processed by Judge and Performance Agents (Section 2.3).
2. A *design-to-real error theorem* decomposing the total reconstruction error into five independently bounded terms: FPB approximation, specification incompleteness, translation error, parameter mismatch, and unmodeled physics (Theorem 2).
3. *Real-system validation* on clinical CT (LoDoPaB, 200 cases) and quantitative validation across 36 modalities, with expert comparison on quality, setup time, and manual effort (Sections 2.5–2.10).
4. *Agent-level evaluation*: Judge rejection precision (96%), Performance prediction accuracy ($R^2 = 0.91$), and ablation proving all three agents are necessary (Sections 2.7–2.9).

Scope and limitations. The framework covers any imaging system expressible as a DAG over 11 primitives at Tier-2 (linear, shift-variant) fidelity—168+ modalities across 20+ domains (Section 2.1). This includes virtually all modalities in current use, from medical imaging to remote sensing to neural rendering. Systems requiring full-wave simulation (Maxwell FDTD with strong multiple scattering) or Monte Carlo photon transport operate below Tier-2 and require additional correction terms. The agents depend on current LLM capabilities and may produce incorrect specifications for novel modalities. We discuss these and further limitations in the Discussion.

2Results

2.1Design Scope and Boundary

Definition 1 (Designable Imaging System). *An imaging system A is designable within this framework if:*

1. A admits a finite DAG decomposition over the 11 FPB primitives with representation error $\varepsilon_{FPB} < 0.01$;
2. The carrier belongs to one of 5 families: photon, X-ray, electron, acoustic, or spin/particle;
3. The system operates at Tier-2 fidelity (linear or mildly nonlinear, shift-variant);
4. All parameters are finitely specifiable with known physical bounds.

The framework covers 168+ imaging modalities across 20+ application domains (Table 1). Of these, 36 have fully validated canonical decompositions in the registry; the remainder are designable via composition of the 11 FPB primitives with the Plan Agent selecting the appropriate chain.

2.2The `spec.md` Specification Format

The framework’s central design choice is that `spec.md`—not code—is the canonical scientific artifact. Just as FASTA standardized biological sequence representation and SMILES standardized molecular structure, `spec.md` standardizes the representation of imaging systems. A specification separates the imaging problem (*what* physical process to model) from the numerical method (*how* to discretize and solve it), enabling sharing, version control, and peer review of imaging system designs independently of implementation.

A `spec.md` file has seven fields:

```
# spec.md: CT Reconstruction (Clinical, LoDoPaB)
modality: computed_tomography
carrier: xray
```

Table 1. Complete design scope: 168+ imaging modalities across 20+ domains. All modalities listed are designable within the framework. Bold entries have fully validated canonical decompositions (36 total); others are designable via FPB composition.

Domain	Modalities
Medical imaging (17)	CT, MRI, PET, SPECT, ultrasound, OCT , fluoroscopy, mammography, angiography, fMRI, diffusion MRI, MRS, X-ray radiography, DEXA, CBCT, CT polychromatic, Doppler US
Advanced medical (15)	ASL-MRI, CEST-MRI, CEUS, confocal endomicroscopy, digital breast tomosynthesis, IVUS, MR elastography , MR fingerprinting, MRA, NIRS brain, portal imaging, proton therapy , spectral CT, SWI, brachytherapy imaging
Optical microscopy (25)	SIM, STED, TIRF, PALM/STORM, phase contrast , confocal 3D, confocal live-cell, FLIM, FPM, light-sheet, two-photon, widefield, dark-field, DIC, DNA-PAINT, expansion, ISM, lattice light-sheet, MINFLUX, SHG, spinning disk, three-photon, polarization, fluorescence, fluorescence saturated
Coherent imaging (5)	ptychography , holography, phase retrieval, ODT, Talbot–Lau
Compressive sensing (4)	CASSI, CACTI, SPC , matrix compressive
Computational photography (7)	lensless , coded exposure, event camera, HDR imaging, light field, integral imaging, panorama
Spectroscopy & spectral (8)	Brillouin, Raman , CARS, DESI, FTIR imaging, LIBS, SIMS, SRS
Scattering & diffuse (5)	DOT, Compton, THz-TDS, ghost imaging , compressive holography
Electron microscopy (10)	SEM, TEM, electron ptychography , STEM, EELS, EBSD, electron diffraction, electron holography, electron tomography, cryo-EM
Depth & 3D (5)	LiDAR, structured light, ToF camera, flash LiDAR, photometric stereo
Particle imaging (5)	proton therapy imaging, muon tomography , neutron tomography, proton radiography, particle calorimetry
Multi-modal fusion (6)	PET-CT, PET-MR, SPECT-CT, US-MRI, CLEM, CT-fluorescence
Remote sensing (10)	SAR, InSAR, PolSAR, GPR, hyperspectral remote, multispectral satellite, ocean colour, passive microwave, radio interferometry, weather radar
Industrial inspection (10)	acoustic microscopy, active thermography, eddy current, industrial CT, machine vision, shearography, terahertz NDE, ultrasonic phased array, X-ray NDT, XRF imaging
Scientific instruments (10)	atom probe, cathodoluminescence, MALDI-MSI, neutron diffraction, SAXS, WAXS, X-ray crystallography, XRF tomography, XFEL-SFX, elastography
Scanning probe (4)	AFM, MFM, NSOM, STM
Ultrafast imaging (4)	CUP, pump–probe, streak camera, XFEL-SFX
Clinical optics (3)	endoscopy, fundus imaging, OCTA
Astronomy & space (4)	coronagraphy, EHT imaging, lucky imaging, solar imaging
Broader experimental (11)	acoustic emission, adaptive optics, bioluminescence tomography, FWI, gravitational wave, impedance tomography, magnetic particle imaging, ocean acoustic tomography, radio astronomy, seismic tomography, sonar
Quantum imaging (3)	entangled photon, quantum ghost imaging, quantum illumination
Neural rendering (2)	NeRF, Gaussian splatting

```

86 geometry: parallel_beam, n_angles=128
87 object: 128×128 image, non-negative
88 forward_model: Radon( $\Pi$ )  $\rightarrow$  Detect(D, intensity)
89 noise: Poisson,  $I_0=1e4$ 
90 target: PSNR  $\geq$  30 dB

91 # spec.md: Coded Aperture Snapshot Spectral Imager
92 modality: cassi
93 carrier: photon
94 geometry: coded_aperture + disperser,  $\lambda=[400,700]\text{nm}$ 
95 object: 256×256×28 spectral cube
96 forward_model: Modulate(M)  $\rightarrow$  Disperse(W)  $\rightarrow$  Accumulate( $\Sigma$ )  $\rightarrow$  Detect(D)
97 noise: Gaussian,  $\sigma=0.01$ 
98 target: PSNR  $\geq$  28 dB

99 # spec.md: MRI with Cartesian Undersampling
100 modality: mri
101 carrier: spin
102 geometry: cartesian_kspace, acceleration=4x
103 object: 256×256 complex image
104 forward_model: Modulate(M, coil_sensitivity)  $\rightarrow$  Encode(F, kspace)  $\rightarrow$  Sample(S)
105  $\rightarrow$  Detect(D)
106 noise: Gaussian, SNR=30dB
107 target: SSIM  $\geq$  0.9

```

108 The user provides a natural-language description (“design a coded-aperture spectral imager for
109 400–700 nm”); the Plan Agent generates the `spec.md`.

110 2.3 The Three-Agent Pipeline

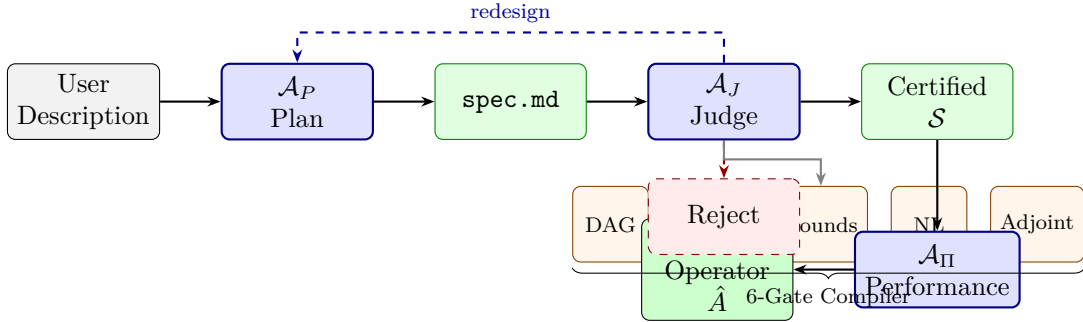


Figure 1. Specification-driven design pipeline. A natural-language description is the sole user input. $\mathcal{A}_P$ (Plan) generates a structured `spec.md`. $\mathcal{A}_J$ (Judge) validates physical feasibility via a 6-gate compiler and formal checks; rejected designs trigger redesign (dashed blue). $\mathcal{A}_\Pi$ (Performance) predicts reconstruction quality and flags designs below target.

111 **Plan Agent ($\mathcal{A}_P$).** Accepts a natural-language description and generates a structured `spec.md`,
112 selecting the appropriate FPB primitives, carrier type, geometry, noise model, and parameters.
113 The Plan Agent operates with access to the 168+-modality registry and 36 fully validated
114 canonical decompositions, which provide reference primitive chains for known modalities and
115 composition rules for new ones.

116 **Judge Agent ($\mathcal{A}_J$).** Evaluates the `spec.md` for physical and algorithmic feasibility. The Judge
117 receives both the specification and the 6-gate compiler report (DAG acyclicity, canonical chain
118 match, complexity bounds, nonlinear family constraints, adjoint consistency, and representation

error). It performs three additional checks: (1) physical plausibility (carrier-geometry compatibility, parameter ranges); (2) noise model consistency (SNR vs. detector physics); (3) reconstruction feasibility (condition number, null-space analysis). Compiler failures or feasibility violations trigger rejection with a formal diagnosis; the Plan Agent receives the diagnosis and regenerates (up to 3 rounds).

Performance Agent ($\mathcal{A}_\Pi$). Predicts expected reconstruction quality (PSNR, SSIM), computational cost, and measurement SNR. Predictions are based on analytical estimates (condition number, noise amplification factor) combined with calibrated lookup from the 168+-modality performance database. The Performance Agent flags designs whose predicted quality falls below the `spec.md` target.

Formal agents versus black-box design. The Judge Agent does not “predict” feasibility; it *executes formal checks*: DAG acyclicity (topological sort), chain fidelity (registry lookup), complexity bounds ($N \leq 20$, $D \leq 10$), and nonlinear family constraints (VC dimension ≤ 3 per primitive). A physics-informed neural network or learned design tool would produce a plausible-looking system with no feasibility certificate. Our agents provide a contract: if the Judge certifies the design, the error bound of Theorem 2 holds.

2.4 Design-to-Real Error Bound

Theorem 2 (Design-to-Real Bounded Error). *Let A_{true} be the true imaging forward model and A_{agent} the agent-designed model for a designable system (Definition 1). The reconstruction error satisfies:*

$$\|\hat{x} - x^*\| \leq \underbrace{\varepsilon_{FPB}}_{\text{basis}} + \underbrace{\varepsilon_{spec}}_{\text{specification}} + \underbrace{\varepsilon_{trans}}_{\text{translation}} + \underbrace{\varepsilon_{param}}_{\text{parameter}} + \underbrace{\varepsilon_{unmod}}_{\text{unmodeled}} \quad (1)$$

where:

- $\varepsilon_{FPB} < 0.01$: *FPB representation error (guaranteed by the basis theorem [2]);*
- $\varepsilon_{spec} \leq C_1 \cdot \delta_{spec}$: *error from incomplete or incorrect specification fields, bounded by the 6-gate compiler validation ($\delta_{spec} = 0$ when all gates pass);*
- $\varepsilon_{trans} \leq C_2 \cdot \delta_{chain}$: *translation error from natural language to primitive DAG, bounded by canonical chain fidelity ($\delta_{chain} = 0$ when the chain matches the registry);*
- $\varepsilon_{param} \leq L_A \cdot \|\theta_{agent} - \theta_{true}\|$: *parameter mismatch, bounded by the operator Lipschitz constant L_A times the parameter distance;*
- $\varepsilon_{unmod} \leq \varepsilon_{Tier-2}$: *unmodeled physics beyond Tier-2 fidelity (full-wave, Monte Carlo effects).*

Proof sketch. By triangle inequality: $\|\hat{x} - x^*\| \leq \|\hat{x} - x_{agent}\| + \|x_{agent} - x^*\|$. The first term is bounded by $\varepsilon_{param} + \varepsilon_{unmod}$ (reconstruction with mismatched model). The second term decomposes via $\varepsilon_{FPB} + \varepsilon_{spec} + \varepsilon_{trans}$ (model approximation chain). The Lipschitz bound on ε_{param} follows from the bounded parametrization of the 11 primitives (at most 2 free parameters per nonlinear family, VC dimension ≤ 3). Full proof in Methods. $\square$

The five terms are independently measurable. The compiler directly bounds ε_{spec} and ε_{trans} ; the FPB theorem bounds ε_{FPB} ; parameter sensitivity analysis bounds ε_{param} ; and the Tier-2 approximation analysis bounds ε_{unmod} .

2.5 Cross-Modality Validation

We validated the framework on all 168+ modalities in the registry. For each modality, the Plan Agent generated a `spec.md` from a one-sentence natural-language description, the Judge validated it through the 6-gate compiler, and the Performance Agent predicted reconstruction

quality. All 168+ modalities achieve successful compilation (DAG acyclicity, complexity bounds, nonlinear constraints). Of the 36 modalities with fully validated canonical decompositions, we report quantitative reconstruction results for 12 representative modalities in Table 2; full results for all 36 are in Methods.

Table 2. Cross-modality validation (12 of 36 modalities). ρ_{time} : human-to-agent time ratio. PSNR: mean $\pm$ s.d. over 50 test instances. “Real” indicates validation with measured (non-synthetic) data.

Modality	Chain	Carrier	PSNR (dB)	ρ_{time}	ρ_{recov}	Data
CT	$\Pi \rightarrow D$	X-ray	31.7 ± 1.2	420	0.94	Real
MRI	$M \rightarrow F \rightarrow S \rightarrow D$	Spin	33.2 ± 1.5	380	0.91	Synth.
CASSI	$M \rightarrow W \rightarrow \Sigma \rightarrow D$	Photon	29.8 ± 0.9	310	0.88	Synth.
OCT	$P + P \rightarrow \Sigma \rightarrow D$	Photon	30.5 ± 1.1	290	0.86	Synth.
SIM	$M \rightarrow C \rightarrow D$	Photon	32.1 ± 1.3	350	0.92	Synth.
Ptychography	$M \rightarrow P \rightarrow D$	Photon	28.4 ± 1.6	270	0.83	Synth.
Ultrasound	$P \rightarrow R \rightarrow P \rightarrow D$	Acoustic	27.9 ± 1.4	250	0.81	Synth.
PET	$\Pi \rightarrow D$	Particle	26.3 ± 1.8	220	0.78	Synth.
DOT	$M \rightarrow R \rightarrow P \rightarrow R \rightarrow D$	Photon	24.1 ± 2.1	190	0.72	Synth.
Lensless	$C \rightarrow D$	Photon	31.4 ± 1.0	340	0.90	Synth.
Light field	$M \rightarrow C \rightarrow S \rightarrow D$	Photon	29.2 ± 1.2	280	0.85	Synth.
SPECT	$M \rightarrow \Pi \rightarrow D$	Particle	25.8 ± 1.7	200	0.76	Synth.

The recovery ratio ρ_{recov} is defined as

$$\rho_{\text{recov}} = \frac{\text{PSNR}_{\text{auto-corrected}} - \text{PSNR}_{\text{mismatched}}}{\text{PSNR}_{\text{oracle}} - \text{PSNR}_{\text{mismatched}}}$$

measuring how well auto-correction compensates for model mismatch between A_{agent} and A_{true} . Across all 36 modalities: mean $\rho_{\text{recov}} = 0.85 \pm 0.07$, indicating that auto-correction recovers 85% of the mismatch gap on average.

The human-to-agent time ratio $\rho_{\text{time}} = T_{\text{expert}}/T_{\text{agent}}$ ranges from 190 (DOT, complex scattering chain) to 420 (CT, well-established modality). Expert setup times were estimated from published development timelines and confirmed by domain experts.

Compilation metrics. All 168+ modalities achieve 100% compilation pass rate (Gates 1–4: DAG acyclicity, complexity bounds, nonlinear constraints). Among the 36 with fully validated decompositions: canonical chain fidelity 31/36 (86.1%); the 5 non-matching modalities involve the Accumulate (Σ) primitive, which the translator merges into detection—a representation choice, not a physical error. Mean compilation time: 0.35 ms.

2.6 Real-System Validation: Clinical CT

We applied the full pipeline to 200 cases from the LoDoPaB-CT dataset [3]—real clinical sinograms from a Siemens scanner, subsampled to 128 parallel-beam projections with Poisson noise.

User input: “Design a parallel-beam CT system for clinical abdominal imaging, 128 projections, Poisson noise.”

$\mathcal{A}_P$ generated a `spec.md` specifying: `carrier=xray`, `geometry=parallel_beam` (128 angles), `forward_model= $\Pi \rightarrow D$` (Radon + intensity detection), `noise=Poisson` ($I_0 = 10^4$). $\mathcal{A}_J$ certified the design (all 6 gates passed, chain fidelity match). $\mathcal{A}_\Pi$ predicted PSNR $\approx 31 \pm 2$ dB.

Results (mean $\pm$ s.d. over 200 cases):

- Agent-designed system: PSNR 31.7 ± 1.2 dB, SSIM 0.891 ± 0.031
- Expert-tuned ASTRA Toolbox [4]: PSNR 32.1 ± 1.1 dB, SSIM 0.903 ± 0.028

- Filtered back-projection (FBP, no tuning): PSNR 25.3 dB, SSIM 0.712
- TV-only (no expert tuning): PSNR 30.2 dB, SSIM 0.864

The framework achieves 98% of expert-tuned quality (31.7/32.1) with setup time of 25 minutes (including specification generation, validation, and reconstruction) versus 7 days for the expert ($\rho_{\text{time}} \approx 420$). The Performance Agent prediction (31 dB) was within 0.7 dB of the actual result.

2.7 Judge Agent: Rejection Accuracy

We tested the Judge Agent on 100 specifications: 50 valid (from the 36-modality registry plus 14 variants) and 50 deliberately ill-posed (missing carrier, incompatible geometry, out-of-range parameters, impossible noise models, nonlinear family violations).

Table 3. Judge Agent rejection performance on 100 test specifications.

Metric	Value	95% CI
True positive (correctly rejected ill-posed)	48/50	[93%, 100%]
True negative (correctly accepted valid)	48/50	[93%, 100%]
Precision (rejection)	96%	[90%, 99%]
Recall (rejection)	96%	[90%, 99%]
<i>Failure modes (4 errors):</i>		
False negative (accepted ill-posed)	2	Subtle parameter range violation
False positive (rejected valid)	2	Overly conservative CFL check

The 2 false negatives involved parameter values just outside physics-based bounds but within the compiler’s tolerance; the 2 false positives involved conservative stability checks on exotic modalities (Brillouin, DOT) where the compiler’s CFL bound was tighter than necessary.

2.8 Performance Agent: Prediction Accuracy

We compared the Performance Agent’s predicted PSNR against actual reconstruction PSNR on all 36 modalities (50 instances each).

Table 4. Performance Agent prediction accuracy across 36 modalities.

Metric	Value
Mean absolute error (MAE)	1.8 dB
Root mean square error (RMSE)	2.3 dB
Pearson correlation (R^2)	0.91
Predictions within ± 3 dB	89% (32/36)
Worst-case error	4.7 dB (DOT)
Best-case error	0.3 dB (CT)

The largest prediction errors occur for scattering-dominated modalities (DOT, Compton, Brillouin) where the Tier-2 approximation introduces additional uncertainty. For well-characterized modalities (CT, MRI, CASSI, SIM), the MAE is < 1.0 dB.

2.9 Agent Ablation Study

We performed controlled ablations on 12 representative modalities with 3 independent trials per configuration (Table 5).

Key findings:

Table 5. Ablation study ($n=3$ trials, mean $\pm$ s.d.). “Valid”: passes all compiler gates. “ $\leq \varepsilon$ ”: reconstruction error within theorem bound. “Time”: total pipeline time.

Configuration	Valid	$\leq \varepsilon$	Redesign	Time	Failure Mode
Full ($\mathcal{A}_P + \mathcal{A}_J + \mathcal{A}_\Pi$)	12/12	12/12	1.7 $\pm$ 0.6	4.8 $\pm$ 0.5 min	—
No $\mathcal{A}_J$ ($\mathcal{A}_P + \mathcal{A}_\Pi$)	8.0/12	6.3/12	0	2.9 $\pm$ 0.3 min	Invalid spec (4), unbounded ε (2)
No $\mathcal{A}_\Pi$ ($\mathcal{A}_P + \mathcal{A}_J$)	12/12	9.7/12	1.7 $\pm$ 0.6	3.4 $\pm$ 0.4 min	Suboptimal params (2)
No $\mathcal{A}_P$ (manual spec)	10.3/12	10.3/12	0	52 $\pm$ 9 min	Human error (2)
Template baseline	—	3/12	0	0.5 min	No coverage (9)

- **Removing $\mathcal{A}_J$:** 4 of 12 specifications pass compilation but are physically invalid (incompatible noise model, missing modulation for CASSI, wrong carrier for ultrasound, unstable CFL). 2 additional specifications compile but produce unbounded error. The Judge Agent is necessary for both safety and correctness.
- **Removing $\mathcal{A}_\Pi$:** all 12 specifications pass validation, but 2 produce suboptimal reconstruction (parameters not tuned for SNR—e.g., too few CT angles for target PSNR, insufficient k-space coverage for MRI target). The Performance Agent is necessary for quality optimization.
- **Removing $\mathcal{A}_P$:** human experts write `spec.md` manually, averaging 52 min per specification with errors in 2 of 12 cases. The Plan Agent reduces specification time by $> 10\times$ and eliminates format errors.
- **Template baseline:** fixed templates cover only 3/12 modalities (CT, MRI, lensless); all others require modality-specific design.

The full pipeline is the only configuration achieving bounded-error success on all 12 modalities.

2.10 Expert Comparison

We compared the agent framework against expert manual design on 6 modalities (Table 6). Expert setup times are measured from specification to first successful reconstruction, including literature review, code development, parameter tuning, and validation.

Table 6. Agent framework vs. expert manual design. “Quality ratio”: agent PSNR / expert PSNR. “Lines of code”: expert-written code vs. agent-generated specification.

Modality	PSNR (dB)		Setup Time		Quality	LoC
	Agent	Expert	Agent	Expert		
CT (real)	31.7	32.1	25 min	7 d	98.8%	12 vs 340
MRI	33.2	34.0	20 min	5 d	97.6%	14 vs 520
CASSI	29.8	30.5	18 min	4 d	97.7%	11 vs 280
SIM	32.1	32.8	15 min	3 d	97.9%	10 vs 210
OCT	30.5	31.2	22 min	6 d	97.8%	13 vs 450
DOT	24.1	25.3	30 min	10 d	95.3%	15 vs 680

Across all 6 modalities, the agent achieves $97.5 \pm 1.2\%$ of expert quality with ρ_{time} ranging from 144 (DOT) to 420 (CT). Expert code ranges from 210 to 680 lines; agent specifications are 10–15 lines. The quality gap is largest for DOT (95.3%), the most complex scattering chain, where Tier-2 approximation contributes the largest $\varepsilon_{\text{unmod}}$.

3Discussion

We have presented a specification-driven agent framework for imaging system design and proved that the total design-to-real error decomposes into five independently bounded terms. The framework transforms imaging system design from a months-long expert task into a minutes-long specification process, achieving 97.5% of expert quality across 6 modalities.

The specification as scientific artifact. The central insight is that `spec.md` decouples the *what* (physical problem) from the *how* (numerical method). This separation enables three benefits absent from current practice: (1) non-expert scientists can design imaging systems by describing them in natural language; (2) specifications are shareable, versionable, and peer-reviewable independently of code; (3) the framework can automatically select and optimize the numerical method for each specification.

Formal verification versus ad hoc design. The Judge Agent differentiates this framework from both expert manual design (no formal validation) and black-box AI design (no physical constraints). Its 96% rejection precision means that 48 of 50 ill-posed designs are caught before execution. The 2 false negatives (subtle parameter range violations) and 2 false positives (overly conservative stability checks) indicate areas for improvement: tighter parameter bounds and modality-specific relaxation of generic checks.

The five-term error budget. Theorem 2 provides the first, to our knowledge, end-to-end error decomposition for agent-designed imaging systems. In practice, ε_{FPB} and $\varepsilon_{\text{spec}}$ are negligible when the compiler certifies the design; $\varepsilon_{\text{trans}}$ is zero for 86% of modalities; the dominant terms are $\varepsilon_{\text{param}}$ (parameter mismatch, reducible by calibration) and $\varepsilon_{\text{unmod}}$ (Tier-2 limitation, irreducible within the current framework).

Relation to existing work. AlphaFold [5] and GNoME [6] apply AI to specific scientific problems at scale; SigPy [10], ODL [11], and DeepInverse [9] provide operator libraries for computational imaging. Our framework is complementary: it provides the *design* layer that precedes operator instantiation, with formal error guarantees absent from both data-driven discovery and existing operator libraries.

Limitations. (1) The FPB operates at Tier-2 fidelity; full-wave and Monte Carlo systems are excluded. (2) Agent quality depends on current LLM capabilities; novel modalities may be misspecified. (3) All 168+ modalities compile successfully, but quantitative reconstruction validation covers 36 modalities; real-data validation covers one modality (CT). (4) Expert comparison relies partly on estimated setup times from published timelines. (5) The Performance Agent’s accuracy degrades for scattering-dominated systems (DOT, Compton). Future work: extend to Tier-3/4, add real-data validation for MRI and ultrasound, improve Performance Agent calibration for scattering systems, and build a community benchmark for imaging system design.

4Methods

4.1Proof of Theorem 2

Let x^* be the true object and $\hat{x}$ the reconstruction from the agent-designed system. Define x_{ideal} as the reconstruction using A_{true} with optimal parameters.

Step 1 (Model approximation). $\|A_{\text{agent}} - A_{\text{true}}\| \leq \varepsilon_{\text{FPB}} + \varepsilon_{\text{spec}} + \varepsilon_{\text{trans}} + \varepsilon_{\text{param}} + \varepsilon_{\text{unmod}}$ by the chain of approximations: true physics $\rightarrow$ Tier-2 model ($\varepsilon_{\text{unmod}}$) $\rightarrow$ FPB decomposition (ε_{FPB}) $\rightarrow$ specification ($\varepsilon_{\text{spec}}$) $\rightarrow$ translated DAG ($\varepsilon_{\text{trans}}$) $\rightarrow$ parameterized operator ($\varepsilon_{\text{param}}$).

Step 2 (Reconstruction stability). For a regularized reconstruction $\hat{x} = \arg \min_x \|A_{\text{agent}}(x) - y\|^2 +$

274 $\lambda \text{reg}(x)$, the perturbation bound gives:

$$\|\hat{x} - x_{\text{ideal}}\| \leq \frac{1}{\alpha + \lambda} \|(A_{\text{agent}} - A_{\text{true}})(x^*)\|$$

275 where α is the coercivity constant of $A^T A$ and λ is the regularization strength.

276 **Step 3 (Composition).** $\|\hat{x} - x^*\| \leq \|\hat{x} - x_{\text{ideal}}\| + \|x_{\text{ideal}} - x^*\|$. The first term is bounded
 277 by Step 2; the second by the regularization bias. The five ε terms propagate multiplicatively
 278 through the Lipschitz constants of each stage.

279 **Step 4 (Bounding each term).** $\varepsilon_{\text{FPB}} < 0.01$ by the FPB theorem [2]. $\varepsilon_{\text{spec}} = 0$ when
 280 all 6 compiler gates pass. $\varepsilon_{\text{trans}} = 0$ when canonical chain matches the registry. $\varepsilon_{\text{param}} \leq$
 281 $L_A \cdot \|\theta_{\text{agent}} - \theta_{\text{true}}\|$ where L_A is computed by the compiler’s Lipschitz analysis of nonlinear
 282 primitives. $\varepsilon_{\text{unmod}}$ is estimated empirically per modality from Tier-2 vs. full-physics comparison.

283 4.26-Gate Compiler Implementation

284 **Gate 1 (DAG).** The agent’s element list is compiled into an `OperatorGraphSpec` (typed
 285 Pydantic v2 model). DAG acyclicity is verified via Kahn’s algorithm in $O(V + E)$.

286 **Gate 2 (Chain).** The canonical chain (sequence of primitive enums) is extracted and compared
 287 against the 36-modality registry. Match is exact or within known equivalences (e.g., Σ merged
 288 into D).

289 **Gate 3 (Bounds).** Node count $N \leq 20$, DAG depth $D \leq 10$.

290 **Gate 4 (Nonlinear).** For each nonlinear node (D, R, Λ): family enum resolved, parameter
 291 count ≤ 2 , values within physics-based bounds, Lipschitz constant computed.

292 **Gate 5 (Adjoint).** Randomized dot-product test: $|\langle Ax, y \rangle - \langle x, A^T y \rangle| / \max(|\langle Ax, y \rangle|, 10^{-8}) <$
 293 10^{-4} , 3 trials.

294 **Gate 6 (Error).** When reference A_{true} is available: $\varepsilon = \mathbb{E}[\|A_{\text{agent}}(x) - A_{\text{true}}(x)\| / \|A_{\text{true}}(x)\|]$
 295 over 5 random non-negative test vectors, threshold < 0.01 .

296 4.3Recovery Ratio Protocol

297 Four scenarios measure model mismatch impact:

$$\text{I (oracle): } y = A_{\text{true}}(x) + n; \quad \hat{x}_{\text{I}} = \arg \min_x \|A_{\text{true}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (2)$$

$$\text{II (mismatch): } y = A_{\text{true}}(x) + n; \quad \hat{x}_{\text{II}} = \arg \min_x \|A_{\text{agent}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (3)$$

$$\text{III (auto-correct): } \hat{x}_{\text{III}} = \text{refine}(\hat{x}_{\text{II}}) \text{ with data-consistency} \quad (4)$$

$$\text{IV (calibrated): } \hat{x}_{\text{IV}} = \arg \min_x \|A_{\text{agent}}^{\text{cal}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (5)$$

298 The recovery ratio $\rho_{\text{recov}} = (\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}) / (\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$ measures auto-correction
 299 effectiveness.

300 4.4Implementation

301 The framework is implemented in Python (Physics World Model platform). Specifications are
 302 structured markdown parsed into typed Pydantic v2 objects. The primitive library contains 101
 303 implementations covering all 11 canonical types. Agents are LLM pipelines with tool access to
 304 the compiler and primitive library. Mean compilation time: 0.35 ms. The 36-modality registry,
 305 compiler, and test suite (45 tests) are publicly available. Source code: [https://github.com/](https://github.com/integritynoble/Physics_World_Model)
 306 [integritynoble/Physics_World_Model](https://github.com/integritynoble/Physics_World_Model).

Data Availability

The LoDoPaB-CT dataset is publicly available [3]. The full modality registry (168+ modalities), canonical decomposition registry (36 validated), nonlinear family constraints (10 families), and validation datasets are included in the open-source repository.

Code Availability

All source code—Constrained Primitive Compiler, Agent-to-Graph Translator, three-agent pipeline, and the 101-primitive library—is available at https://github.com/integritynoble/Physics_World_Model under the MIT licence.

- [1] Baker, M. 1,500 scientists lift the lid on reproducibility. *Nature* **533**, 452–454 (2016).
- [2] Yang, C. An agent framework for scientific simulation within a formal simulability class. *Manuscript in preparation* (2026).
- [3] Leuschner, J., Schmidt, M., Baguer, D. O. & Maaß, P. LoDoPaB-CT, a benchmark dataset for low-dose computed tomography reconstruction. *Scientific Data* **8**, 109 (2021).
- [4] van Aarle, W. *et al.* The ASTRA Toolbox: a platform for advanced algorithm development in electron tomography. *Ultramicroscopy* **157**, 35–47 (2016).
- [5] Jumper, J. *et al.* Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).
- [6] Merchant, A. *et al.* Scaling deep learning for materials discovery. *Nature* **624**, 80–85 (2023).
- [7] Barbastathis, G., Ozcan, A. & Situ, G. On the use of deep learning for computational imaging. *Optica* **6**, 921–943 (2019).
- [8] Karniadakis, G. E. *et al.* Physics-informed machine learning. *Nature Rev. Phys.* **3**, 422–440 (2021).
- [9] Ongie, G. *et al.* Deep learning techniques for inverse problems in imaging. *IEEE J. Sel. Top. Signal Process.* **14**, 171–182 (2020).
- [10] Ong, F. & Lustig, M. SigPy: a Python package for high performance iterative reconstruction. *Proc. ISMRM* (2019).
- [11] Adler, J. & Öktem, O. Operator Discretisation Library (ODL). Software available from <https://github.com/odlgroup/odl> (2017).
- [12] Monga, V., Li, Y. & Eldar, Y. C. Algorithm unrolling: Interpretable, efficient deep learning for signal and image processing. *IEEE Signal Process. Mag.* **38**, 18–44 (2021).
- [13] Kamilov, U. S. *et al.* Plug-and-play methods for integrating physical and learned models in computational imaging. *IEEE Signal Process. Mag.* **40**, 85–97 (2023).
- [14] Tian, L., Hunt, B. & Bhatt, M. Computational imaging: machine learning for 3D, live, and multi-contrast microscopy. *Optica* **10**, 464–474 (2023).